Physics - Undergraduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is
   (a) mc/2
   (b) mc/($2^{(}$1/2))
   (c) mc
   (d) ($3^{(}$1/2))mc

2. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
   (a) Planck's constant
   (b) Boltzmann's constant
   (c) The Rydberg constant
   (d) The speed of light

3. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R. The time required for one revolution is
   (a) independent of M
   (b) proportional to $m^{(}1/2)$
   (c) linear in R
   (d) proportional to $R^{(}3/2)$

4. A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is
   (a) 4
   (b) 5
   (c) 6
   (d) 20

5. A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?
   (a) 0.25 W
   (b) 0.5 W
   (c) 1 W
   (d) 4 W

True / False (10 marks)

Answer True or False

6. True or False: In a reversible thermodynamic process, the internal energy of the system changes significantly.
7. True or False: Electromagnetic radiation emitted from a nucleus is often in the form of visible light.
8. True or False: The speed of light in a nonmagnetic dielectric material with a dielectric constant of 4.0 is 1.5 x 10^8 m/s.
9. True or False: The Hall coefficient can be used to determine the sign of the charge carriers in a doped semiconductor.
10. True or False: The negative muon, mu^{-}, $haspropertiesthataremostsimilartothoseoftheelectron$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the similarities between the emission spectra of the doubly ionized lithium atom Li++ and hydrogen, and analyze how the wavelengths of Li++ are affected by its atomic structure.
12. A tube of water is moving at 1/2 c relative to a lab frame. Analyze the behavior of a light beam entering the tube and calculate its speed relative to the lab frame, considering the index of refraction.

End of Paper